EXP 9: Compare Linear and Polynomial Regression

CODE

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.metrics import r2_score
X = np.array([1, 2, 3, 4, 5, 6]).reshape(-1, 1)
y = np.array([2, 5, 10, 17, 26, 37])
linear = LinearRegression()
linear.fit(X, y)
linear_pred = linear.predict(X)
poly = PolynomialFeatures(degree=2)
X_poly = poly.fit_transform(X)
polynomial = LinearRegression()
polynomial.fit(X_poly, y)
poly_pred = polynomial.predict(X_poly)
print("Linear R2 Score:", round(r2_score(y, linear_pred), 3))
print("Polynomial R2 Score:", round(r2_score(y, poly_pred), 3))
plt.scatter(X, y, color='blue', label='Actual Data')
plt.plot(X, linear_pred, color='red', label='Linear Regression')
plt.plot(X, poly_pred, color='green', marker='o',
        label='Polynomial Regression')
plt.xlabel("X Values")
plt.ylabel("Y Values")
plt.title("Linear vs Polynomial Regression")
plt.legend()
plt.grid()
plt.show()
```

OUTPUT

Linear R2 Score: 0.958

Polynomial R2 Score: 1.0

